

Notes: Becker & Milbourn (JFE, 2011)

How did increased competition affect credit ratings?

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1 Big picture

- **Question.** Does increasing competition among credit rating agencies improve ratings quality?
- **Natural experiment.** The paper studies the rise of Fitch as a credible third competitor to Moody's and Standard & Poor's in the U.S. corporate bond ratings market.
- **Industry setting.** Ratings are issuer-paid, publicly used by investors, and deeply embedded in regulation and private contracting. This makes ratings quality economically important.
- **Treatment variable.** Competition is measured as Fitch's market share in a two-digit NAICS industry-year cell: the fraction of corporate bond ratings issued by Fitch rather than Moody's or S&P.
- **Empirical variation.** Fitch's growth was large but uneven across industries. This cross-industry variation is the key empirical lever.
- **Quality dimension 1: rating levels.** Higher incumbent ratings can indicate lower quality because ratings categories lose stable meaning when inflated.
- **Quality dimension 2: market informativeness.** If ratings are informative, they should be strongly related to bond yield spreads.
- **Quality dimension 3: default prediction.** If ratings classify risk well, they should predict future corporate defaults.
- **Main result.** Increased Fitch competition is associated with lower quality ratings from Moody's and S&P: ratings rise, the rating-yield relation weakens, and ratings predict default less well.
- **Magnitude.** A one-standard-deviation increase in Fitch market share is predicted to raise average ratings by about 0.19 notches in a baseline firm-rating specification, roughly one rating-step upgrade for one out of five firms.
- **Default-prediction result.** Speculative-grade firms are much more likely than investment-grade firms to default when Fitch competition is low, but the gap is much smaller when competition is high.
- **Mechanism interpretation.** The evidence is consistent with reputation models in which future rents motivate current ratings quality. More competition lowers incumbent rents and weakens incentives to preserve reputation.
- **Policy takeaway.** Competition may reduce oligopoly rents, but in an issuer-paid ratings market it may also reduce ratings quality. The usual pro-competition intuition is therefore incomplete.

2 Data and measurement

2.1 Why credit ratings are economically important

- Ratings summarize default and recovery risk for investors who may not perform their own detailed credit analysis.
- Regulations and private contracts rely on ratings categories; for example, many institutional investors face investment-grade constraints.
- The issuer-paid business model creates a conflict: investors want accurate ratings, while issuers prefer favorable ratings that lower financing costs.

Interpretation. The key question is whether competition pushes raters to produce better information, or instead pushes them to please issuers by inflating ratings.

2.2 Fitch as the new competitor

- Fitch had old roots, but became a significant competitor after recapitalization, merger with IBCA in 1997, and acquisitions of Duff & Phelps and Thomson Bankwatch in 2000.
- Fitch’s market share increased particularly quickly after 2000 and varied substantially across industries.
- In July 2005 Fitch was added to the Lehman bond index classification rule; after this, the middle of Moody’s, S&P, and Fitch ratings mattered for index classification.

Interpretation. The rise of Fitch gives intertemporal and cross-industry variation in ratings competition.

2.3 Competition measure

For industry j in year t , define

$$FitchShare_{jt} = \frac{\# \text{ ratings issued by Fitch in industry } j, t}{\# \text{ ratings issued by Fitch, Moody’s, and S\&P in industry } j, t}.$$

- The baseline uses two-digit NAICS industries from 1995 to 2006.
- The paper also uses moving averages of Fitch market share for figures.
- The industry-level measure is intended to capture competition faced by incumbents, not whether a particular bond has a Fitch rating.

Interpretation. This distinction matters because bond-level Fitch presence can reflect selection or ratings shopping, while industry-level market share captures the competitive environment.

2.4 Ratings data

- Firm ratings, default events, and accounting data come from Compustat Industrial and Operating Segments databases.
- Corporate bond ratings are mapped to a numerical scale running from 28 for AAA to 1 for D.
- Bond ratings are observed repeatedly over time; the paper controls for lagged ratings and fixed effects where appropriate.

Interpretation. The numerical scale allows regressions of ratings levels, but the authors also use ordered probit and collapsed cells to reduce dependence on cardinality assumptions.

2.5 Bond yield data

- Bond transactions come from Mergent FISD insurance-company trades.
- The paper excludes bonds with features that complicate yield calculations, including floating-rate, callable, puttable, convertible, exchangeable, foreign-currency, and Yankee bonds.
- The dependent variable in yield tests is the yield spread to the closest-maturity Treasury bond.
- The sample contains a little more than 110,000 bond transactions after restrictions.

Interpretation. Bond prices provide a market benchmark for the information content of ratings. If ratings are informative, higher ratings should imply lower yield spreads.

2.6 Default data

- The default-prediction tests use firm-year observations and study whether the firm defaults within three years.

- For data reasons, the default tests use S&P issuer ratings.
- Defaults are rare, especially for high-rated firms, so the tests are demanding.

Interpretation. Default prediction is the cleanest risk-classification test, but rare defaults limit statistical power.

2.7 Controls

The regressions include combinations of:

- year fixed effects and industry fixed effects;
- firm fixed effects in some firm-rating specifications;
- industry-by-year or industry-year interaction fixed effects in robustness tests;
- firm controls such as size, assets, cash, EBITDA, leverage, interest expense, PPE, and sales;
- bond controls such as maturity, offering amount, issue timing, and previous rating.

Interpretation. The controls are designed to separate competition from changes in firm credit quality, industry demand for ratings, and bond-specific features.

3 Models and empirical specifications

3.1 Ratings inflation test

The baseline rating-level specification is

$$Rating_{ijt} = \alpha + \beta FitchShare_{jt} + \Gamma' X_{ijt} + \delta_j + \tau_t + \varepsilon_{ijt}.$$

- $Rating_{ijt}$ is a firm or bond rating on the numerical scale.
- $FitchShare_{jt}$ is the competition measure.
- X_{ijt} contains firm or bond controls.
- δ_j and τ_t are industry and year fixed effects, with alternative specifications using firm fixed effects.

Interpretation. A positive β means that incumbents issue more favorable ratings when Fitch competition is stronger. The paper treats this as ratings inflation and therefore lower classification quality.

3.2 Bond-rating test with Fitch presence

At the bond level, the paper estimates variants of

$$BondRating_{bjt} = \alpha + \beta FitchShare_{jt} + \phi FitchRated_{bjt} + \Gamma' X_{bjt} + FE + u_{bjt}.$$

Interpretation. Including *FitchRated* addresses selection: bonds that obtain a Fitch rating may differ from those that do not. The key result is that industry-level Fitch share remains positive even when bond-level Fitch presence is negative.

3.3 Leverage-interaction test

To test whether issuers that care more about ratings benefit more from competition, the paper estimates

$$Rating_{ijt} = \alpha + \beta_1 (FitchShare_{jt} \times Leverage_{it}) + \beta_2 Leverage_{it} + \Gamma' X_{it} + FE + \varepsilon_{ijt}.$$

Interpretation. A positive interaction means that rating inflation is stronger for more levered firms, consistent with raters placing more weight on issuer preferences when competition rises.

3.4 Industry-demand and entry checks

A generic version of the entry check is

$$FitchShare_{jt} = \alpha + \rho Z_{j,t+k} + \delta_j + \tau_t + u_{jt},$$

where Z includes the number of ratings issued, industry debt, changes in industry debt, and industry profitability, using leads and lags.

Interpretation. If Fitch entered industries because credit demand or profitability was changing, these variables should predict Fitch market share. The paper finds little evidence for this.

3.5 Yield-spread informativeness test

The yield-spread specification is

$$Spread_{bjt} = \alpha + \beta_1 Rating_{bjt} + \beta_2 FitchShare_{jt} + \beta_3 (Rating_{bjt} \times FitchShare_{jt}) + \Gamma' X_{bjt} + FE + \varepsilon_{bjt}.$$

- Since higher ratings mean safer bonds, $\beta_1 < 0$ is expected.
- If ratings become less informative when competition is high, the rating–yield relation weakens, so $\beta_3 > 0$.

Interpretation. This test asks whether market prices treat ratings as less meaningful in high-Fitch-share industries.

3.6 Default-prediction test

The default specification is

$$Default3yr_{it} = \alpha + \beta_1 Rating_{it} + \beta_2 FitchShare_{jt} + \beta_3 (Rating_{it} \times FitchShare_{jt}) + \Gamma' X_{it} + FE + \varepsilon_{it}.$$

Interpretation. Higher ratings should predict lower default probability. A positive interaction term means this predictive relation weakens when Fitch competition is higher.

3.7 Instrumental variables

The IV design instruments actual Fitch share using predicted Fitch share based on 1995 market share and the 2006 median market share:

$$\widehat{FitchShare}_{jt} = FitchShare_{j,1995} + \frac{t - 1995}{2006 - 1995} (0.353 - FitchShare_{j,1995}).$$

Interpretation. The instrument relies on the idea that Fitch grew faster in industries where it initially had low share, while the 1995 share is predetermined relative to later rating-quality outcomes.

4 Identification and empirical design

4.1 What identifies the main effect

- Identification comes from differences across industries and over time in Fitch’s market share.
- Year fixed effects absorb aggregate rating trends.
- Industry fixed effects absorb stable differences across sectors.
- Firm controls and firm fixed effects absorb observable and time-invariant issuer differences.

Interpretation. The maintained assumption is that, conditional on controls and fixed effects, changes in Fitch competition are not driven by unobserved shocks that independently lower incumbent rating quality.

4.2 Main endogeneity concerns

- **Ratings shopping:** issuers may obtain Fitch ratings when Fitch is favorable.
- **Reverse causality:** Fitch may enter industries where ratings are already high or where incumbents are weak.
- **Credit-demand shocks:** Fitch may grow where debt issuance and rating demand are expanding.
- **Information opacity:** Fitch may enter sectors where default is intrinsically harder to predict.

Interpretation. The paper does not claim randomized entry. It uses several tests to rule out the most natural alternative explanations.

4.3 Why ratings shopping does not explain the main result

- Moody's and S&P essentially rate all taxable corporate bonds issued in the United States, including unsolicited ratings.
- Individual bonds rated by Fitch tend to have lower, not higher, incumbent ratings after controls.
- The aggregate result goes the other way: industries with greater Fitch share have higher incumbent ratings.

Interpretation. Selection into Fitch ratings appears to bias against the paper's main result rather than explain it.

4.4 Why industry-demand tests matter

- The paper checks whether Fitch share is related to the number of ratings issued, industry debt, changes in industry debt, and profitability.
- These tests use leads and lags of the demand proxies.
- The evidence shows no systematic relation between Fitch growth and obvious credit-market demand variables.

Interpretation. This reduces concern that Fitch simply entered booming or deteriorating sectors where ratings would have changed anyway.

4.5 Why yield and default tests matter

- Ratings levels alone could reflect different standards, but weaker relation to yields shows that ratings contain less market-relevant information.
- Weaker default prediction shows that ratings also classify risk less well.
- These are complementary notions of ratings quality.

Interpretation. The paper's strongest evidence is the joint pattern across all three quality measures: level, yield informativeness, and default prediction.

4.6 What the paper cannot fully prove

- Fitch market share is not randomly assigned.
- The IV strategy relies on the validity of the 1995-share-based predicted market share.
- The paper focuses on corporate ratings, not structured products.

- The mechanism is inferred from theory and patterns in the data rather than directly observing internal rating-agency effort or issuer bargaining.

Interpretation. The evidence strongly supports a reputation-rents interpretation, but the paper remains an observational study.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* style. Nearby figures and tables are grouped to save space and improve comparison. The visuals are directly cropped from the original paper.

5.1 Figures 1–2: Fitch’s evolution and market-share growth

Read:

- Figure 1 shows the corporate history behind Fitch’s rise: IBCA merger in 1997 and Duff & Phelps/BankWatch acquisitions in 2000 were central.
- Figure 2 shows Fitch market share rising from below 10% in the median industry in the late 1990s to around 30% by 2007.
- The 25th and 75th percentile lines show large cross-industry dispersion, which the empirical design exploits.

Economic message. Fitch’s growth created a real change in the competitive structure of the corporate ratings market.

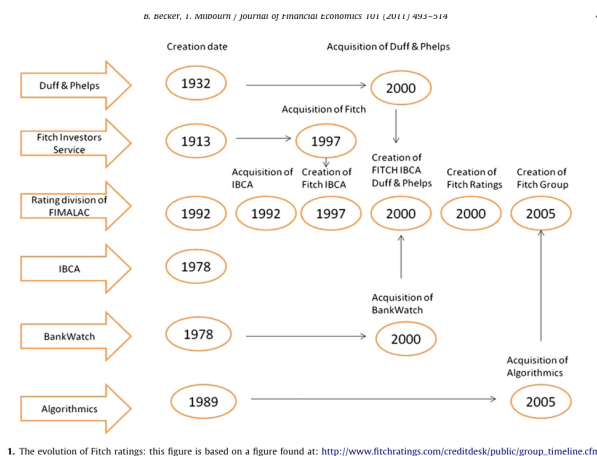


Figure 1: evolution of Fitch Ratings

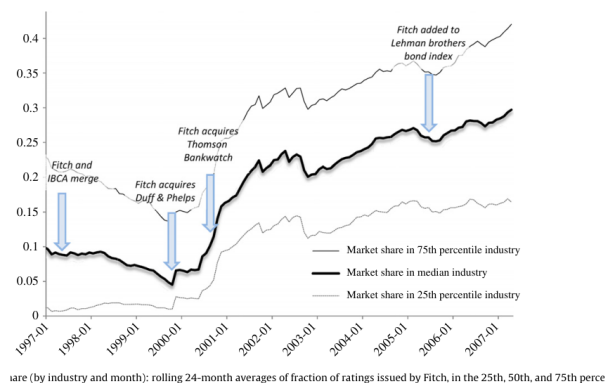


Figure 2: Fitch market share by industry and month

5.2 Tables 1–2: market-share variation and ratings scale

Read:

- Table 1 shows that Fitch’s market share rises in almost all industries from 1995–2000 to 2001–2006.
- Industries with high late-sample Fitch presence include real estate, retail trade, finance/insurance, utilities, management of companies, and public administration.
- Table 2 maps Moody’s, S&P, and Fitch ratings to the numerical scale used in the regressions.
- Investment-grade ratings correspond to BBB/Baa and above, while below that are speculative-grade categories.

Economic message. The paper converts qualitative ratings into a numerical risk scale and uses industry-level Fitch share as the continuous competition measure.

Table 1
Fitch market share by two-digit North American Industry Classification System (NAICS) industry early and late in sample.
Each row presents the straight average of annual market shares for the earlier and later half of the sample period, respectively. Market share is the fraction of instrument ratings for corporate bonds issued by firms in a two-digit NAICS industry that have been issued by Fitch, as opposed to Moody's and Standard & Poor's.

NAICS	Industry name	Fitch market share	
		1995–2000	2001–2006
11	Agriculture, Forestry, Fishing and Hunting	0.09	0.23
21	Mining, Quarrying, and Oil and Gas Extraction	0.22	0.26
22	Utilities	0.29	0.34
23	Construction	0.08	0.28
31	Manufacturing: Food, Textile, Apparel	0.17	0.30
32	Manufacturing: Wood, Paper, Printing, Petroleum, Chemicals, plastics	0.16	0.26
33	Manufacturing: Metals, Machinery, Computers, Electrical, Furniture	0.18	0.28
42	Wholesale Trade	0.15	0.24
44	Retail Trade: Motor Vehicles, Furniture, Electronics, Food, Gas	0.09	0.28
45	Retail Trade: Sporting goods, Books, Florists, Office Supplies, Mail-Order, Vending	0.20	0.40
48	Transportation and Warehousing: Air Transport, Water Transport, Trucks, Pipelines	0.12	0.15
49	Transportation and Warehousing: Messengers, Storage	0.06	0.28
51	Information	0.18	0.28
52	Finance and Insurance	0.30	0.36
53	Real Estate and Rental and Leasing	0.17	0.42
54	Professional, Scientific, and Technical Services	0.18	0.27
55	Management of Companies and Enterprises	0.17	0.34
56	Administrative and Support and Waste Management and Remediation Services	0.12	0.29
61	Educational Services	0.11	0.28
62	Health Care and Social Assistance	0.16	0.27
71	Arts, Entertainment, and Recreation	0.16	0.21
72	Accommodation and Food Services	0.13	0.30
81	Other Services (except Public Administration)	0.12	0.22
92	Public Administration	0.28	0.36

Table 2
The ratings scale.
The table describes categories for credit ratings, as well as the numerical scale used in the paper. Multiple numerical values for a single rating level represents the number assigned to ratings with a + qualifier, no qualifier, and a – qualifier, respectively. The source for ratings definitions is Standard & Poor's (S&P) Ratings Definitions from March 17, 2008 (Standard and Poor, 2009).

Rating group	Rating agency		Numerical value assigned	Category definition
	Moody's	S&P, Fitch		
Investment grade	AAA	AAA	28	The obligor's capacity to meet its financial commitment on the obligation is extremely strong.
	Aa	AA	24, 25, 26	The obligor's capacity to meet its financial commitment on the obligation is very strong.
	A	A	21, 22, 23	Somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than obligations in higher-rated categories. However, the obligor's capacity to meet its financial commitment on the obligation is still strong.
	Baa	BBB	18, 19, 20	Exhibits adequate protection parameters. However, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity of the obligor to meet its financial commitment on the obligation.
Default	Ba	BB	15, 16, 17	Obligations rated 'BB', 'B', 'CCC', 'CC', and 'C' are regarded as having significant speculative characteristics. 'BB' indicates the least degree of speculation and 'C' the highest. While such obligations will likely have some quality and protective characteristics, these may be outweighed by large uncertainties or major exposures to adverse conditions.
	B	B	12, 13, 14	
	Caa	CCC	9, 10, 11	
	Ca	CC	7	
Default	C	C	4	
	D	D	–	An obligation in payment default. The 'D' rating category is used when payments on an obligation are not made on the date due even if the applicable grace period has not expired, unless Standard & Poor's believes that such payments will be made during such grace period. The 'D' rating also will be used upon the filing of a bankruptcy petition or the taking of a similar action if payments on an obligation are jeopardized.

5.3 Table 3 and Figure 3: sample statistics and rating distribution

Read:

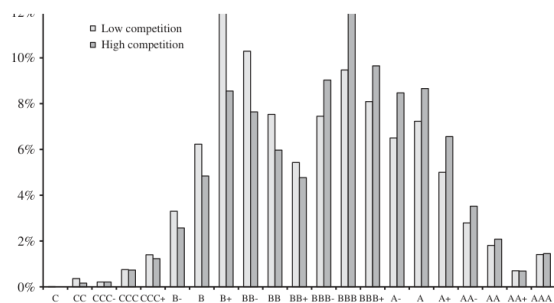
- Table 3 reports mean Fitch market share of 0.212 and standard deviation of 0.142.
- The bond-rating sample is very large: 686,990 bond credit-rating observations.
- The yield-spread sample has 110,955 bond trades.
- Figure 3 shows that investment-grade ratings are more common in high-competition industries and junk ratings are more common in low-competition industries.

Economic message. The raw distribution already points toward ratings inflation under higher Fitch competition.

Table 3
Summary statistics.
Each column presents the coefficient estimates from an ordinary least squares or logistic specification. Intercepts are not reported. The sample period is from 1995 until 2006. The left-hand-side variable is coded as follows: AAA=28, AA=26, A=24, A-23, A+=25, BBB=18, BB=16, BB+=15, B=14, B+=13, CCC=11, CC=7, C=4. Firm characteristics are measured at the end of the previous fiscal year (using accounting data from Compustat). Leverage is debt over total assets. Yield spread is the yield to maturity minus the yield of the closest maturity treasury bond. BBTDs average before interest, taxes, depreciation, and amortization.

	Firm credit rating	Bond credit rating	Fitch market share	Yield spread	Yield spread non-investment grade	Yield spread non-investment grade	Yield spread at issue	Leverage	BBTD	Default in three years
Mean	18.082	21.080	0.212	194.6	128.4	530.1	164.0	0.368	3.798	0.010
SD	18	21	0.235	121.3	119.3	467.3	188.0	0.341	2.840	0.009
Standard deviation	1.939	4.943	0.142	252.7	139.9	390.8	158.3	0.269	4.073	0.101
Number of observations	10,750	686,990	429	1,110,955	92,940	19,008	7,510	10,750	10,300	18,051

Source: Fitch IBCA, 2007.



3. Firm credit ratings distribution: high and low competition (Fitch market share above and below median).

5.4 Tables 4–5: ratings levels rise with Fitch competition

Read:

- Table 4 shows that Fitch market share positively predicts firm credit ratings across OLS, ordered probit, firm fixed-effect, and collapsed-cell specifications.
- In column (2), the coefficient is 1.325; with a one-standard-deviation Fitch-share increase of 0.142, average ratings rise by about 0.19 notches.
- Table 5 shows a similar positive relation for individual bond ratings.
- The Fitch presence dummy in Table 5 is negative, meaning bonds that themselves receive Fitch ratings have lower incumbent ratings, which works against the main industry-level result.

Economic message. Incumbents issue more favorable ratings when Fitch competition is stronger, and this cannot be explained by simple bond-level selection into Fitch ratings.

Table 4
Predicting firm credit ratings with Fitch market share.
Each column presents the coefficient estimates from an ordinary least squares (OLS) or ordered probit specification. Intercepts are not reported. The sample period is from 1995 until 2006. The left-hand-side variable refers to credit opinion ratings by Standard and Poor's and is coded from 28 (AAA) to 1 (D). See Table 4 for details. Fitch market share is the fraction of bond ratings in an industry-year cell performed by Fitch Ratings. Firm characteristics are the log of sales, log of book value of assets, cash divided by total assets (and its square), EBITDA (earnings before interest, taxes, depreciation, and amortization) divided by total assets (and its square), cash flow over total assets (and its square), EBITDA over sales (and its square), cash flow over sales (and its square), PPE (property, plant, and equipment) over total assets (and its square), interest expense over EBITDA (and its square), and debt over total assets (and its square), all measured at the end of the previous fiscal year (using accounting data from Compustat). Industries are two-digit level North American Industry Classifications System (NAICS) industries. In Columns 5 and 6, data are collapsed by industry-year cell (averages and medians, respectively). Heteroskedasticity-robust standard errors for the coefficient estimates are in parentheses. Errors are clustered by industry $\times$ year cell in Columns 1 to 4. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	Dependent variable: firm credit rating					
	OLS (1)	OLS (2)	OLS (3)	Ordered probit (4)	OLS average by cell (5)	OLS median by cell (6)
Fitch market share	2.395** (1.123)	1.325** (0.566)	0.784* (0.432)	0.3615** (0.156)	1.533** (0.570)	1.754** (0.846)
Year fixed effects	No	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	No	Yes	No	Yes	Yes	Yes
Firm fixed effects	No	No	Yes	No	No	No
Firm controls	No	No	Yes	No	Yes	Yes
R ²	0.004	0.141	0.900	n/a	0.961	0.914
Number of observations	19,630	19,630	19,630	19,630	160	160

Table 5
Predicting bond ratings with Fitch market share.
Each column presents the coefficient estimates from an ordinary least squares (OLS) regression. Intercepts are not reported. The sample period is from 1995 until 2006, except in Columns 5 and 6, which refer to 1996–2006. The left-hand-side variable refers to credit opinion ratings by Standard and Poor's and is coded from 28 (AAA) to 1 (D). See Table 4 for details. For Column 1 to 4, the unit of observation is an individual bond rating. For Columns 5 and 6, the unit of observation is industry-year cell, and all variables are the averages (Column 5) or medians (Column 6) of all observations in that cell. Cells with less than ten bond ratings are excluded. Fitch market share is the fraction of bond ratings in an industry-year cell performed by Fitch Ratings. Fitch rating (dummy) is a dummy variable taking the value one if Fitch issued a rating for the bond issue in the same calendar year as the rating was made. Industries are two-digit level North American Industry Classifications System (NAICS) industries. Previous rating refers to the same bond issue's preceding rating. Heteroskedasticity-robust standard errors for the coefficient estimates are in parentheses. Errors are clustered by industry $\times$ year cell. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	Dependent variable: bond issue credit rating					
	OLS, all ratings (1)	OLS, all ratings (2)	OLS, all ratings (3)	OLS, excludes NAICS 52 (4)	OLS, average, collapsed by cell; excludes NAICS 52 (5)	OLS, medians, collapsed by cell; excludes NAICS 52 (6)
Fitch market share	0.4187** (0.187)	0.5024** (0.220)	0.5941*** (0.198)	0.6793** (0.336)	2.061* (1.214)	5.201*** (1.782)
Fitch presence (dummy)			–0.0938*** (0.045)			
Maturity					0.0312 (0.088)	–0.205* (0.118)
Lagged rating (collapsed)					0.332*** (0.095)	0.242** (0.095)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	No	Yes	No	Yes	Yes
Time to maturity fixed effects	Yes	Yes	Yes	Yes	No	No
Previous rating fixed effects	Yes	Yes	Yes	No	Yes	No
Bond issue fixed effects	No	Yes	No	Yes	No	No
R ²	0.941	0.959	0.941	0.904	0.908	0.842
Number of observations	368,811	368,811	368,811	146,366	176	176

5.5 Tables 6–7: leverage heterogeneity and entry checks

Read:

- Table 6 shows that competition-induced ratings inflation is stronger for more levered firms.
- The interactions with leverage, long-term leverage, high-leverage dummy, and debt/EBITDA are all positive and statistically significant.
- Table 7 checks whether Fitch market share is driven by rating demand, industry debt, changes in industry debt, or profitability.
- Most lead, lag, and simultaneous coefficients are insignificant.

Economic message. Ratings inflation is stronger where issuers care more about ratings, while obvious industry demand variables do not explain Fitch entry.

Table 6
Predicting rating levels with Fitch market share–interactions with leverage.
Each column presents the coefficient estimates from an ordinary least squares (OLS) or logistic specification. Intercepts are not reported. The sample period is from 1995 until 2006. The left-hand-side variable refers to firm credit opinion ratings by Standard and Poor's and is coded from 28 (AAA) to 1 (D). See Table 4 for details. Fitch market share is the fraction of bond ratings in an industry-year cell performed by Fitch Ratings. Firm characteristics are the log of sales, log of book value of assets, cash divided by total assets (and its square), EBITDA (earnings before interest, taxes, depreciation, and amortization) divided by total assets (and its square), cash flow over total assets (and its square), EBITDA over sales (and its square), cash flow over sales (and its square), PPE (property, plant, and equipment) over total assets (and its square), interest expense over EBITDA (and its square), debt over total assets (and its square), all measured at the end of the previous fiscal year (using accounting data from Compustat). Leverage is debt over total assets, long term leverage is long-term debt over assets, the high leverage dummy is equal to one if debt over assets is above 0.2324 (the sample median). Industries are two-digit level North American Industry Classifications System (NAICS) industries. The standard errors for the coefficient estimates are in parentheses and are clustered by industry $\times$ year cell. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	Dependent variable: firm credit rating			
	OLS (1)	OLS (2)	OLS (3)	OLS (4)
Fitch market share $\times$ leverage	6.010*** (1.645)			
Fitch market share $\times$ long-term leverage		5.317*** (1.708)		
Fitch market share $\times$ high leverage dummy variable			1.587** (0.682)	
Fitch market share $\times$ debt/EBITDA				0.3806*** (0.129)
Debt/EBITDA				0.3221*** (0.0483)
Firm controls	Yes	Yes	Yes	Yes
Industry $\times$ year fixed effects	Yes	Yes	Yes	Yes
R ²	0.589	0.599	0.588	0.629
Number of observations	19,630	19,630	19,630	19,630

Table 7
Fitch's market share and segment characteristics.
Each coefficient estimate refers to one ordinary least squares (OLS) specification (different rows represent regressions that differ only in the timing of the independent variable). Each regression includes year and industry fixed effects. For each regression, the coefficient estimate for Fitch's market share is reported. The sample period is from 1995 until 2006. Number of ratings issued is the log of the aggregate number of credit ratings issued for bonds in an industry. Industry debt is the log of the total amount of outstanding debt of Compustat firms in an industry. The change in industry debt is the log of industry debt minus its previous value. Industry profitability is an asset-weighted average of the ratio of EBITDA (earnings before interest, taxes, depreciation, and amortization) to assets in Compustat firms in the industry and industry profitability (median) is the median EBITDA-asset ratio across all Compustat firms in the industry. Fitch market share is the fraction of bond ratings in an industry-year cell performed by Fitch Ratings. The number of observations is 266 or fewer (some observations are lost due to lags). The standard errors for the coefficient estimates are in parentheses and are heteroskedasticity-robust. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

Timing of Fitch's market share	Independent variable: Fitch's market share				
	Number of ratings issued (1)	Industry debt (dollars) (2)	Change in industry debt (dollars) (3)	Industry profitability (4)	Industry profitability (median) (5)
Lead ($t+2$)	0.1831 (0.4055)	–0.1079 (0.1937)	–0.0885 (0.2039)	–0.0108 (0.0272)	0.0085 (0.0156)
Lead ($t+1$)	0.0194 (0.4210)	0.0824 (0.1913)	0.2870 (0.1794)	–0.0167 (0.0248)	0.0039 (0.0155)
Simultaneous (t)	0.1474 (0.4151)	–0.0617 (0.1739)	–0.1813 (0.1541)	–0.0086 (0.0251)	0.0112 (0.0158)
Lag ($t+1$)	0.2968 (0.3367)	–0.2615 (0.1673)	–0.1362 (0.1584)	0.0151 (0.0245)	–0.0008 (0.0149)
Lag ($t+2$)	0.2015 (0.3190)	–0.3075* (0.1768)	0.1018 (0.1577)	0.0213 (0.0237)	0.0221 (0.0165)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	≤ 266	≤ 266	≤ 266	≤ 266	≤ 266

5.6 Table 8: bond-yield informativeness

Read:

- Credit rating has a negative coefficient: better ratings are associated with lower yield spreads.
- The interaction of credit rating with Fitch market share is positive in every specification.

- This means the negative relation between ratings and yield spreads becomes weaker when Fitch competition is higher.
- The effect appears for all trades, investment-grade bonds, noninvestment-grade bonds, and initial-issue spreads.

Economic message. Ratings become less informative about bond prices when incumbent rating agencies face more Fitch competition.

Table 8

Bond yields and ratings—the effect of Fitch market share.

Each column presents the coefficient estimates from an ordinary least squares (OLS) regression. Intercepts are not reported. Each observation is the yield to maturity of a bond in one transaction. The sample period is from 1995 until 2006. The dependent variable is the yield to maturity minus the yield to maturity of the government bond with the closest maturity. Credit ratings are bond credit ratings issued by Standard and Poor's and Moody's (reported by the Margent Fixed Income Securities Database), and represent the latest preceding the transaction (if several were issued simultaneously, we use the average), not older than three months. Fitch market share is the fraction of bond ratings in an industry-year cell issued by Fitch Ratings. Industries are two-digit level North American Industry Classifications System (NAICS) industries. Bonds are excluded if they have nonstandard features (see text for details), negative yields, or yields above 20%. The standard errors for the coefficient estimates are in parentheses and are clustered by industry $\times$ year cell. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	Dependent variable: yield spread				
	OLS, all trades (1)	OLS, all trades (2)	OLS, investment grade only (3)	OLS, noninvestment grade only (4)	OLS, initial issue (5)
Credit rating $\times$ Fitch market share	0.766*** (0.162)	0.968*** (0.214)	0.430*** (0.114)	1.253** (0.523)	0.430*** (0)
Credit rating	−0.569*** (0.162)	−0.620*** (0.063)	−0.272*** (0.003)	−1.102*** (0.164)	−0.287*** (0.034)
Fitch market share	−14.160** (3.544)				
Log of time to maturity	−1.174*** (0.168)	−1.203*** (0.165)	−1.121*** (0.125)	−3.414*** (0.493)	0.666*** (0.085)
Log of time to maturity, squared	0.251*** (0.035)	0.268*** (0.035)	0.278*** (0.027)	0.495*** (0.115)	−0.113*** (0.022)
Log of offering amount	−0.223** (0.101)	−0.302*** (0.108)	0.366*** (0.071)	0.665 (1.581)	−0.260 (0.352)
Log of offering amount, squared	0.002 (0.004)	0.006 (0.004)	−0.019*** (0.003)	−0.018 (0.063)	−0.007 (0.015)
Date fixed effects (month-year)	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	No	No	No	No
Year-industry fixed effects	No	Yes	Yes	Yes	Yes
R ²	0.484	0.503	0.444	0.398	0.697
Number of observations	110,955	110,955	92,949	18,006	7,427

5.7 Table 9: default prediction

Read:

- Table 9 shows that ratings predict default less well when Fitch competition is higher.
- The investment-grade dummy interaction with Fitch share is positive, so the default gap between investment-grade and speculative-grade firms shrinks with competition.
- The rating-by-Fitch-share interaction is also positive in the numerical-rating specifications.
- The paper reports that speculative-grade firms are 7.7 times as likely to default as investment-grade firms at low competition, but only 2.2 times as likely at high competition.

Economic message. The classification meaning of ratings deteriorates when competition increases.

Table 9

Default prediction—the effect of Fitch market share.

Each column presents the coefficient estimates from an ordinary least squares (OLS) regression. Intercepts are not reported. Each observation is one firm-year in which firm-level controls can be identified and the firm is identified as defaulting or not defaulting in three years. The sample period is from 1995 until 2005. Fitch market share is the fraction of bond ratings in an industry-year cell issued by Fitch Ratings. Industries are two-digit level North American Industry Classifications System (NAICS) industries. Firm characteristics are the log of sales, log of book value of assets, cash divided by total assets (and its square), EBITDA (earnings before interest, taxes, depreciation, and amortization) divided by total assets (and its square), cash flow over total assets (and its square), EBITDA over sales (and its square), cash flow over sales (and its square), PPE (property, plant, and equipment) over total assets (and its square), interest expense over EBITDA (and its square), debt over total assets (and its square), and the log of sales and the log of assets, all measured at the end of the previous fiscal year (using accounting data from Compustat). In Column 5, data are averaged by industry-year cell. The standard errors for the coefficient estimates are in parentheses and are clustered by industry $\times$ year cell in Column 1 to 4 and heteroskedasticity robust in all columns. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	Default in three years, OLS (1)	Default in three years, OLS (2)	Default in three years, OLS (3)	Default in three years, OLS (4)	Fraction default in three years by industry-year cell, OLS (5)
IG dummy $\times$ Fitch market share	0.089*** (0.030)				
IG dummy	−0.033*** (0.067)				
Rating $\times$ Fitch market share		0.0123*** (0.070)	0.0123*** (0.081)	0.0063* (0.0038)	0.0070*** (0.0001)
Rating		−0.0045*** (0.0009)	−0.0049*** (0.0011)		−0.0094*** (0.004)
Fitch market share	−0.080*** (0.028)	−0.253*** (1.250)	−0.229*** (0.085)	−0.116 (0.080)	−0.121*** (0.023)
Year fixed effects	No	No	Yes	Yes	Yes
Industry fixed effects	No	No	Yes	Yes	Yes
Year and industry FE $\times$ rating	No	No	No	Yes	No
Firm controls	No	No	Yes	No	Yes
R ²	0.008	0.001	0.024	0.024	0.571
Number of observations	18,707	18,650	15,661	18,650	189

Table 10

Predicting bond and firm ratings with Fitch market share—instrumental variables.

Each column presents the coefficient estimates from an ordinary least squares (OLS) or ordered probit specification. Intercepts are not reported. The sample period is from 1996 until 2006. The left-hand-side variable refers to credit opinion ratings by Standard and Poor's and is coded from 28 (AAA) to 1 (D). See Table 4 for details. Fitch market share is the fraction of bond ratings in an industry-year cell performed by Fitch Ratings. Fitch rating (dummy) is a dummy variable taking the value one if Fitch issued a rating for the bond issue in the same calendar year as the rating was made. Industries are two-digit level North American Industry Classifications System (NAICS) industries. Previous rating refers to the same bond issue's preceding rating. The standard errors for the coefficient estimates are in parentheses and are clustered by industry $\times$ year cell. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

Second stage	Dependent variable: firm credit rating			
	2SLS, all ratings (1)	2SLS, all ratings (2)	2SLS, all ratings (3)	2SLS, all ratings (4)
Fitch market share	5.7475** (1.941)	14.290 (10.011)	6.074*** (2.254)	11.771*** (2.299)
Instrumented)				
Year fixed effects	No	Yes	Yes	Yes
Industry fixed effects	No	Yes	Yes	Yes
Firm controls	No	No	No	Yes
Firm fixed effects	No	No	Yes	Yes
First stage				
Predicted market share	0.8815*** (0.0107)	0.3419*** (0.0189)	0.2962*** (0.0242)	0.2010*** (0.0245)
First stage R ²	0.289	0.719	0.744	0.772
Second stage R ²	0.003	0.108	0.882	0.900
Number of observations	16,810	16,810	16,810	16,810

5.8 Table 10: instrumental-variable robustness

Read:

- The instrument predicts Fitch share using initial 1995 market share and extrapolation toward the 2006 median market share.
- First-stage coefficients are strong and highly significant.
- IV estimates are generally larger than OLS estimates; columns (1), (3), and (4) are statistically significant.
- The IV results support the view that measurement error or reverse causality is not driving the main ratings-inflation result.

Economic message. The IV evidence reinforces the central result: increased competition from Fitch is associated with more favorable incumbent ratings.

6 Short summary

- The paper studies whether more competition improves credit-rating quality.
- The setting is the rise of Fitch as a significant third competitor to Moody's and S&P in corporate bond ratings.
- Competition is measured as Fitch's market share by two-digit NAICS industry-year.
- Ratings quality is evaluated using ratings levels, relation to bond yields, and default prediction.
- Increased Fitch market share is associated with higher Moody's and S&P ratings.
- The ratings increase is stronger for more levered firms, where issuers care more about ratings.
- Individual bonds rated by Fitch tend to have lower incumbent ratings, so simple ratings shopping does not explain the aggregate industry-level result.
- The relation between ratings and bond yield spreads weakens when Fitch market share is high.
- Ratings also become worse at predicting defaults when Fitch market share is high.
- Industry demand, debt growth, and profitability do not systematically explain Fitch market share.
- Instrumental-variable tests based on 1995 Fitch market share support the main result.
- The evidence is most consistent with reputation models: more competition reduces expected future rents and weakens incentives to maintain quality.
- The policy implication is that competition in ratings markets may reduce rents but may not improve information quality.

7 Conclusion

7.1 Core conclusion

Becker and Milbourn find that the rise of Fitch as a third competitor did not improve incumbent ratings quality. Instead, higher Fitch market share is associated with higher ratings, lower ratings informativeness for bond yields, and weaker default prediction.

7.2 Economic intuition

Ratings are produced in a market where issuers pay, investors use ratings for free, and reputational capital disciplines the agencies. When competition reduces future rents, the value of maintaining a costly reputation

can fall. Incumbents may then put more weight on issuer preferences, leading to more favorable but less informative ratings.

7.3 Takeaway

The paper is a warning against simple pro-competition policy in information markets. In markets where quality is hard to observe and reputation is sustained by future rents, competition can reduce quality rather than improve it.